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Development of a multi-dimensional measure for assessing entrepreneurial ecosystems

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ABSTRACT

Researchers and theorist have put great effort into defining and examining entrepreneurial ecosystems and how business clusters develop in certain regions. Favourable entrepreneurship ecosystems are thought to drive business and innovation. However, a commonly accepted measure of entrepreneurial ecosystem favourableness has yet to be developed. The purpose of the present research is to contribute to ecosystems research by taking a two-study approach to developing and validating a perceptual measure of entrepreneurial ecosystems. The perceptual measure is based upon prior conceptual frameworks that outline ecosystems. In developing this measure, we are able to further unpack and illuminate the factor structure of ecosystems, the results of which have direct scholarly and practitioner uses.

KEYWORDS

Entrepreneurship ecosystems; clusters; innovation; measurement; scale development

Introduction

Clusters of innovative firms emerge under certain conditions (Isaksen 2016; Oerlemans, Meeus, Boekema 2001; Uyarra and Ramlogan 2016) and many antecedents dictate success and failure of firms, including firm innovation, within such clusters. For example, geographical aspects of local ecosystems affect firm innovation (Herstad 2017). We refer to these as entrepreneurial ecosystems (EE), which we define as conditions that make ecosystems more or less favourable for entrepreneurship activity. According to the EE perspective, firms cluster in advantageous regions and thrive on various environmental resources such as information, networks and labour markets (Coenen, Moodysson, and Martin 2015).

Research in this space includes topics such as ecosystem entrepreneurs (Nambisan and Baron 2013; Velt, Torkkeli, and Saarenketo 2018), social and sustainable entrepreneurship ecosystems (McMullen 2018; Muñoz and Dimov 2015), entrepreneurial university ecosystems (Hayter 2016; Shaeffer and Matt, 2016) and location-based innovative clusters for high growth and other geographic implications (Breschi and Malerba 2001; Li et al. 2016), all of which help us better understand where firms may cluster to innovate and thrive. However, although conceptual attempts to clarify how ecosystems are measured (e.g. Isenberg 2010a, 2011a, 2011b), empirical evidence still lags behind. The purpose of this study is to better define and measure entrepreneurship ecosystems through the development of a Multidimensional Entrepreneurial Ecosystem Scale (MEES) capable of assessing six specific domains within an entrepreneurial ecosystem. In doing so, we help to unpack the elements of geographical regions that encourage firm clustering and innovation.

EE researchers no longer surround questioning if territory or environment are important elements, yet now acknowledge the importance of territory and environment as undeniable; in doing so, the research has shifted towards understanding and validating the effectiveness and efficiency of environmental configurations (Boutillier, Carre, and Lavrato 2016). Conceptually, an EE is a collection of six non-causal critical domains of entrepreneurship. These include policy, finance, culture, supports, human capital and markets (Isenberg 2011b). Within each of these six domains reside many sub-domains. For example, within policy reside leadership and government, each has sub-domain items. Culture includes success stories and societal norms, which also have sub-domain items. Not of surprise, the market domain includes early customers, but it also includes networks. Isenberg (2011a) provides a more comprehensive overview. Given new products, new jobs and economic growth in the United States are largely a result of the entrepreneurial sector (Hisrich and Grachev 1993; Baumol and Strom 2007), and considering that many economies have changed drastically in the last fifteen years due to technology and globalization, the study of EE is prudent to better understand what drives entrepreneurship and what can make a region more or less suitable for sustaining an entrepreneurial cluster.

A number of indices and barometers exist, each purportedly measuring unique aspects of the EE at one level or another. For example, efforts such as the Global Economic Monitor (GEM) and Doing Business (DB) help capture EE aspects at the national level, enabling cross-national comparison. Yet, these efforts are difficult to scale down to a more local level. The National Federation of Independent Business's National Small Business Poll (NSBP), designed to capture a snapshot of the United States as a whole, is more easily adapted to the state or regional level, but lacks some of the robustness of GEM and DB. In the United States, popular press outlets such as Site Selection Magazine and Development Consultants International offer metrics to rank state level ecosystems (e.g. Guttman 2015; Stamer 2005). Yet, a critical review of these metrics call their validity into question given they typically use small, non-representative convenience samples that lack any theoretical foundation (e.g. Arend 2004; Newman 2015).

Therefore, we seek to address these issues and contribute to the literature by developing a new, individual level, multidimensional measure of an EE: the MEES. Using Isenberg's conceptual work (2010a, 2011a), we provide needed systemization of the six domains of the entrepreneurial ecosystem (policy, finance, culture, supports, human capital and markets) as a framework. A fundamental contribution of the MEES is that it can be used to assess ecosystems at any level; all the researcher must do is define the scope of the ecosystem when employing the measure. Thus, MEES can look at ecosystems at a city, county, state, region, or national level. Second, we tap into ecosystems as a cognitively assessed, rather than objectively assessed, construct. We believe this is a strength of our study as it is perceptions that influence behaviours and 'perceptions can be more discriminating than objective measures' (Powell 1996, 336). We accomplish our goal by assessing policy, finance, culture, supports, human capital and markets through the perception of individuals within the given ecosystem. Fourth, we argue that understanding which characteristics lead to entrepreneurship is of upmost importance and that too few efforts have been made to understand the broader context. Lastly, given entrepreneurship and clustering is of critical importance to economic growth and innovation (Herstad 2017; Wennekers and Thurik 1999), having an accurate measurement tool provides a great number of practical contributions (i.e. what conditions are conducive to entrepreneurship).

The remainder of the manuscript is as follows: we first address current measures of entrepreneurship ecosystems. Next, we provide a more in-depth overview of our guiding principles and the six domains of the MEES. Third, we describe our item generation procedure. This is followed by Study 1, Study 2 and the results of our analyses. Lastly, we discuss the results, implications, limitations and opportunities for future research.

Current measures of entrepreneurial ecosystems

The existing literature is dominated by three means to assess entrepreneurial ecosystems (viz., GEM, PSED, and Kauffman), each of which we will discuss further. We will not explore the popular

press approaches noted earlier given their lack of scientific rigour. First, the Global Entrepreneurship Monitor (GEM) is arguably the largest and most long-standing option, allowing for ‘...a deep understanding of the environment for entrepreneurship’ (GEM Consortium, 2017 para. 5). To date, GEM studies have been conducted in over 100 countries and involve over 300 academic and research institutions. GEM, however, is primarily focused at the national level, with past GEM research resulting in the development of ‘country profiles’ or comparing differences in nations, not individual communities. Second, the Panel Study of Entrepreneurial Dynamics (PSED), was ‘designed to enhance the scientific understanding of how people start businesses’ and data are collected on the process of business formation from nationally representative samples. Included in the PSED are some environmental impact items hitting at some of the core aspects of the entrepreneurial ecosystem. Like GEM, this measure has been used primarily at the national level (Parker and Belghitar 2006; Reynolds and Curtain 2009), but unlike GEM the focus is more on the person and process and less on the environment.

Third, the Kauffman Foundation has published a variety of research reports and guides to assessing and exploring entrepreneurial ecosystems. We draw specific attention to their March 2015 white paper titled ‘Measuring an Entrepreneurial Ecosystem’ by Dane Stangler and Jordan Bell-Masterson. Stangler and Bell-Masterson (2015) posit that while some attempts to measure entrepreneurship include only one or two indicators, others try to include the kitchen-sink approach and suggest there should be some common ground to better understand this puzzle.

Together, GEM, PSED and the Kauffman Foundation research reports have been the largest and most influential sources of insight to those who study entrepreneurial ecosystems. Yet, none of the three enables policy makers and other interested parties to fully nor adequately assess the favourableness of the ecosystem at the community level and from the perspective of the constituent. For this reason, we chose to create the MEES.

Creating a new instrument: MEES

Given many past measures have either been specific to the national level, rooted solely in economic indicators, or having been ad hoc to study only a small subset of phenomenon, we sought to develop a more comprehensive measure of the entrepreneurial ecosystem within a given community. Isenberg (2011b) posited a six-domain taxonomy for considering entrepreneurial ecosystems at the local level: policy, finance, culture, supports, human capital, and markets. Using these six domains, and two guiding principles, the MEES was developed.

Guiding principle #1 – applicability

First, we began with the contention that this measure should be broad enough to work in any geography, yet specific enough to capture the uniqueness of a given locality. Consider Fresno, California for example. Fresno has a growing entrepreneurial ecosystem, especially in regard to technology and agriculture (e.g. Fallows 2015; Santos 2018). Yet, despite this activity, few would call Fresno a major tech or start-up hub. Now consider Silicon Valley, the ‘gold standard’ of technology. Both Fresno and Silicon Valley are California cities, subject to California labour laws, California tax and legal structures, and California geography; all factors that constrain either cities ability to differ from one another across some major domains. Despite standardized state and federal labour laws, state income tax rates, and west coast location, Fresno suffers from a severe brain drain problem (i.e. an inability to effectively retain its most promising human capital) while Silicon Valley arguably leads the nation in human capital.

Given that firms seek out knowledge-based human capital (Coder, Peake, and Spiller 2017; Roper, Love, and Bonner 2017), this presents an interesting regional difference. While these two locations vary in a very small amount from a policy perspective, they vary greatly in culture, access to markets, and human capital, all of which have the ability to impact growth (Li et al. 2016).

Thinking across the United States one can quickly identify numerous cities that vary across multiple domains despite them both residing within the same nation, region, and state (Austin vs. Houston; Nashville vs. Memphis; Miami vs. Pensacola; etc.). Isenberg (2010a) suggested that to build a successful entrepreneurial ecosystem one must embrace and exploit the local conditions of the specific community rather than trying to emulate what works in other communities (viz., stop trying to replicate Silicon Valley), so a tool enabling a locality to better understand its specific environmental context is needed.

Guiding principle #2 – attitudes affect behaviour

The second guiding principle included is that behaviour is best predicted by intent (Armitage and Conner 2001). Given intent is heavily derived from attitudes (Ajzen 2001), understanding how a particular community views a given domain is critical to our understanding of each domain, especially at the local level. While individual views may be informed by objective reality, bounded rationality and information asymmetry are at play. Considering the finance domain, for example, a steadily increasing number of SBA loan applications approved or the cumulative dollar total of SBA lending within a community rising over a 5-year period may not equate to the community recognizing improved access to capital. If there is a general sentiment within a community that banks will not lend, there will be fewer loan applications received. In sum, we presume that understanding the intentions of the community at large is critical to fully understanding the strengths and weaknesses of the community's ecosystem.

Domain #1 – policy

Policy is the extent to which government and leadership not just support and advocate for entrepreneurial activity, but also have rules and regulations in place to govern it. At the national level this could be the difference between the intellectual property protection systems of the United States vs. Asia; at the state level this could be the difference between tax and entity structures in Texas vs. California; and at local level this could be the permitting, zoning, and occupational licensing processes in place that vary from city to city and county to county. Regulation that impacts venturing is prevalent given most countries have experienced impactful regulation increases over the past century (e.g. Kleiner 2000; Kleiner and Krueger 2010).

Domain #2 – finance

Finance centres primarily around access to capital. This includes not just the presence of venture capitalists, angel investors, and traditional bank lenders, but also state and community run micro-loan programmes (e.g. Benjamin, Rubin, and Zielenbach 2004). Given 67.2 per cent of new ventures are financed from founder savings (Kauffman 2015), finance also includes the wealth of the individuals in the community (this impact can be profound given friends and family account for another 28.4% per cent of new venture financing nationally; Kedrosky and Stangler 2011).

Domain #3 – culture

Organizational researchers have long recognized the importance of culture in stimulating creativity and innovation (Martins and Terblanche 2003) and affecting organizational performance (Den Hartog and Verborg 2004). Moreover, research conducted by the Organization for Economic Cooperation and Development (OCED) found that entrepreneurial culture is a key determinant of EE success (Boutillier, Carré, and Levratto 2016). EE Culture is largely individually driven by people advocating values that promote innovation and venturing as viable career opportunities (Audretsch 2007; Hechavarria and Ingram 2018). Mack and Mayer (2016) showed the impact

success story visibility has in building culture. Others work has shown how a tolerance for risk and acceptance of failure are important cultural characteristic. For example, up until the 1980s, Ireland was heavily stigmatized by failure (viz., loan default or bankruptcy) to the point where children to coached take corporate or government jobs and not start new ventures, but some successes in the 1990s caused a change in how failure is viewed in Ireland and enterprising has become much more socially acceptable (Isenberg 2010a).

Domain #4 – supports

Supports for entrepreneurship go a long way to enabling and empowering a community to truly facilitate entrepreneurial behaviour. Supports include infrastructure (transportation; high-speed Internet access; energy), support professions (legal, accounting), and entrepreneurship friendly institutions and programmes (small business resource centres, chambers of commerce, business plan competitions). Tampa, FL is a good example of how supports can both help and hinder the entrepreneurial ecosystem. Tampa has 62 entrepreneurial support organizations (White et al. 2016), access to accounting and legal expertise, the promise of Google Fibre internet infrastructure (Trigaux 2015), and a plethora of entrepreneurial programming open to the community (Startup Weekends, Startup Week, co-working spaces, Small Business Development Centers). All of these factors contribute to Tampa's continual rise in numerous rankings related to entrepreneurship. But, Tampa also has a fatal infrastructure flaw that hinders its ecosystem: transportation. Tampa has poor public transportation and the ability to connect the main research and technology hubs of the city with the core business community is often cited as a major impediment to the Tampa EE (Kritzer 2016). Similarly, research into Estonia's EE has found infrastructure is a hindrance to their growth and prosperity (Kshetri 2014; Velt, Torkkeli, and Saarenketo 2018; fortunately, this same work notes that Estonia's culture is especially strong and responsible for the boom of entrepreneurial activity the country has experienced in the last 15 years).

Domain #5 – human capital

Access to human capital has long been a factor impacting entrepreneurial venturing (Bendickson et al. 2017; Huggins, Prokop, and Thompson 2017). Whereas in some industries access to human capital has become less of an issue given technological advances (e.g. videoconferencing, cloud-based collaboration tools, etc.), other industries still very much rely on the availability of adequate human capital in their geographic vicinity making this a critical EE success factor. Randall Kemper, Executive Director of the Aspen Network of Development Entrepreneurs, notes that access to human capital is an especially impactful hindrance for many for smaller companies in emerging markets (NewsGhana 2016). Moreover, even in developed markets access to adequate human capital regularly affects both venture growth and success (Siepel, Cowling, and Coad 2017).

Domain #6 – markets

Markets include access to early adopters, distribution channels, diaspora networks, etc. South Korea serves as a good example of how access to markets can help or hinder the EE. South Korean entrepreneurs rely heavily on diaspora-based networks in the United States and Europe to not just access resources but also develop out their markets (Kitching, Smallbone, and Athayde 2009). Similarly, major U.S. companies test new products in cities that serve as microcosms to the nation as a whole (e.g. Nashville, Orlando, Cincinnati, etc.; Pilny 2014).

Item generation

Following systematic item generation procedures (cf. Netemeyer, Bearden, and Sharma 2003; Hinkin 1995, 1998), several sources were scoured for potential items that could reflect the six posited domains. These sources included the Global Entrepreneurship Monitor, Doing Business, the National Federation of Independent Business' National Small Business Poll, the Panel Study for Entrepreneurial Dynamics, Feld's (2012) Startup Communities, and the entrepreneurship literature (cf., Fietters, Greene, and Rice 2010; Hitt et al. 2011; Isenberg 2010a, 2010b, 2011a, 2011b; Isserman 1993; Schwarzkopf 2016; Venkataraman 1997).

Then, following a deductive approach (Schwab 1980; Spector 1992), additional items were developed based upon semi-structured interview discussions with four subject matter experts spanning various business domains (e.g. accounting, finance, marketing, management). Each expert was provided with a standardized definition of an entrepreneurial ecosystem from Isenberg (2010b). The lead author then asked the interviewees about what characteristics the expert considered to be the most impactful indicators of a strong entrepreneurial ecosystem. Once their responses were recorded the experts were shown the six-dimension model and asked what indicators they would expect to see in their community under strong and weak domain conditions.

To ensure the breadth of coverage was not limited to academics or the academic literature, two full-time Small Business Development Center Counsellors who interact regularly with nascent entrepreneurs were also consulted and given an opportunity to help generate items from the entrepreneur's perspective. Lastly, to ensure no age or culture biases exist, two diverse focus groups ($N = 80$; 74% Caucasian, $\mu = 21.54$ years of age; 55% female) of undergraduate students at a large research university were conducted to help the research team generate items. The use of students was limited only to item generation and much prior research substantiates the relevance of considering adults at 21 years of age or higher as potential entrepreneurs (Turker and Selcuk 2009) given they have an increased propensity for new venturing (Linan and Santos 2007).

In addition to the item generation procedures followed above, three expert judges, each familiar with both psychometric theory and the entrepreneurial ecosystem literature, were solicited and agreed to assist in purifying the item pool following Haynes, Richard, and Kubany's (1995) content validation guidelines. Working independently, judges were asked to review each item for clarity, meaningfulness, and appropriateness. Judges were provided the latitude to rephrase or eliminate items they deemed redundant, ambiguous, misleading or poorly worded. Likewise, they were also empowered to write or suggest new items. This process resulted in 43 useable items.

Sample 1

Two independent samples were used to develop and validate the MEES. Following recommendations from Prescott and Soeken (1989) to pre-test the measure as a means of sharpening the design of a proposed study prior to formal data collection, we collected data (viz., Sample 1) from a snowball sample of nascent entrepreneurs across the United States ($N = 198$). Sample 1 spanned the Northeast, Southeast and West Coast. Sample 1 was used to explore the basic factor structure of the MEES and its correlates.

Sample 2

Next, to confirm and validate the MEES, we surveyed entrepreneurs in a mid-sized metropolitan city on the West Coast (viz., Sample 2). Respondents for Sample 2 were recruited using a targeted sampling approach allowing for access to multiple organizations (Schneider 1987; Schneider, White, and Paul 1998; Organ and McFall 2004; Organ, Podsakoff, and MacKenzie 2006) across multiple industries ($N = 267$). Respondents represented industries including restaurants, hospitality, financial services, construction, technology, retail trade and healthcare.

Validity testing

Following recommendations from Netemeyer, Bearden, and Sharma (2003) we explored both translation- and criterion-related validity. The first, translation validity, was established via the item generation and development process employed. The second, criterion-related validity is evidenced via convergent validity. More specifically, across both samples the internal consistency reliability estimates for each domain were at or above the generally accepted 0.70 level, and the items, by and large, loaded cleanly on the theoretically hypothesized domain.

Results

Exploratory factor analysis

To reduce the original set of items down to a number of items that best reflected each dimension of the MEES, we used an exploratory factor analysis (EFA) technique on our sample of nascent entrepreneurs (i.e. sample 1). EFA is a dependence technique that categorizes items together based on how strongly they load onto a common factor (Pedhazur and Schmelkin 1991). Specifically, we performed six common factor analyses (viz., one factor per theorized dimension) thus forcing the analysis to settle upon a single factor solution for each of the analyses in order to identify items that did not fit its respective domain well. Items were then eliminated on the basis of poor factor loadings (i.e. we eliminated items that displayed loadings of less than .7 until all items within each factor had loadings of greater than .7, or the set of items representing each factor was limited down to a set of three¹). This process yielded a set of 22 items that can be expected to consistently tap the dimensions of people's perceptions of a given entrepreneurial ecosystem. See Table 1 for a list of the items and their respective factor loadings that were obtained through this process. Note that when analyzed individually each factor analysis yielded loadings that greatly exceed the .5 cut-off value required for practical significance (Hair et al. 2006).

Additionally, all of the loadings were significant at the .05 level. The factor solution also maintained its structure when all of the items were entered into the analysis together and analyzed as a group, see Table 2.² That is, the items representing each respective factor consistently loaded together and there were no meaningful or problematic dual loadings. Although, some of the loadings within this final EFA were rather low, the loadings do hover around the .5 cut-off level (i.e., .49–.8) suggested for practical purposes such as data reduction (Hair et al. 2006). Furthermore, all of the factor loadings were significant at the .05 level.

Confirmatory factor analysis

After reducing the data down to a smaller set of items that were most reflective of the focal constructs, the data from the sample of entrepreneurs (sample 2) were then analyzed through the use of confirmatory factor analysis (CFA). CFA was selected as the ideal method to test the structure of our data, because it allows the researcher to test the extent to which their hypothesized factor model fits the data. Specifically, CFA is used to test how well the hypothesized factor model is able to reproduce the observed covariances between a set of items. Hence, in the present study CFA allowed us to test how well the proposed factor structure of the MEES fits the actual data. We estimated a model that contained all six dimensions of the MEES with each set of respective items loaded onto their corresponding dimension to test our hypothesized model.

Upon the first run of the CFA with the set of 22 items identified during the EFA we yielded an inadequate fitting model that did not meet the standard cut-off value for model fit in terms of incremental fit. In addition, the root mean square error of approximation (RMSEA) value was also too large at first run. We recognized that a number of poor fitting (viz., items with loadings of less than .7 within factors that had more than three items) within the factors of *Supports* and *Finance* were the root cause of this problem. Specifically, we found that items 3 and 4 within the *Supports*

Table 1. Summary of individual EFAs.

Factor	Loading
Finance	
Finance_2: There are local individual investors in my community who are willing to financially support entrepreneurial venturing.	0.82
Finance_5: Bankers in my community work hard to help entrepreneurs obtain financing.	0.76
Finance_3: Financing for entrepreneurship is available in my local community.	0.76
Finance_4: Information on what funding programmes are available for entrepreneurs is easily accessible.	0.74
Finance_1: My community has a sufficient number of banks who are willing to lend to entrepreneurs.	0.72
Supports	
Supports_2: My community has the infrastructure necessary to start and run most businesses (e.g. telecommunication, transportation, energy).	0.77
Supports_3: My community has many entrepreneur-friendly organizations such as Rotary Clubs or Chambers of Commerce.	0.76
Supports_6: Professional services (e.g. lawyers and accountants) for entrepreneurs are readily available in my community.	0.75
Supports_5: I believe the resources in my community are well designed to support business growth.	0.74
Supports_4: Local organizations, such as incubators and Small Business Development Centers, are active in supporting local entrepreneurs.	0.71
Culture	
Culture_17: The social values and culture of the community emphasize creativity and innovativeness.	0.85
Culture_16: The social values and culture of the community encourage entrepreneurial risk-taking.	0.85
Culture_15: The social values and culture of the community emphasize self-sufficiency, autonomy, and personal initiative.	0.77
Human Capital	
Human Capital_3: Local educational institutions offer specialized courses in entrepreneurship.	0.87
Human Capital_4: There are entrepreneurial training programs, such as entrepreneurship bootcamps, available in my local community.	0.80
Human Capital_2: There are ample local institutions of higher education (universities, community colleges, trade schools) within my community.	0.80
Markets	
Markets_7: The diversity in my community provides a great test market for many other locations.	0.84
Markets_6: My community networks could help me distribute new products across a variety of new markets.	0.83
Markets_5: My community's multinational diversity helps keep me connected the global economy.	0.82
Policy	
Policy_4: The local government actively seeks to create and promote entrepreneurship-friendly legislation.	0.89
Policy_3: The local government has programmes in place to help new entrepreneurs, such as seed funding programmes or entrepreneurship training programmes.	0.79
Policy_5: Local community leaders regularly advocate for entrepreneurs.	0.67

*Sample size $N = 198$. All factor loadings are significant at $p < .05$

factor displayed a low set of standardized loadings and items 4 and 5 within the *Finance* also displayed a similarly low set of standardized loadings. Therefore, we rejected the four poor fitting items and ultimately yielded a set of 18 items to represent the six factors of the MEES (three items per factor). See [Figure 1](#) for an overview of the final factor structure.

We assessed model feasibility and validity through the use of MPLUS as well as via a number of model fit indices in order to demonstrate goodness-of-fit, incremental (or comparative) fit, and a lack of badness-of-fit (Hair et al. 2006). See [Table 3](#) for a summary of the model fit statistics. First, we used the chi-square goodness-of-fit statistic to assess overall model fit. However, the classic chi-square goodness-of-fit statistic is heavily influenced by sample size, that is, tests of model fit involving samples of even moderate sizes (e.g. $N = 100$ – 400) are often rejected as poor fitting (i.e. $P < .05$) (Marsh, Balla, and McDonald 1988). Hence, the fact that our data sample and CFA model failed this test was a rather predictable and non-problematic result. Therefore, to provide an additional test of model fit that is able to compensate for larger sample sizes, we also used the normed-chi-square goodness-of-fit statistic. The normed-chi-square goodness-of-fit statistic is useful because it is a statistic based on dividing the goodness-of-fit statistic through by the number of degrees of freedom, to discount for model parsimony (Kline 2005). Additionally, the CFI was used to test for incremental model fit and the RMSEA was used to test for badness-of-fit.

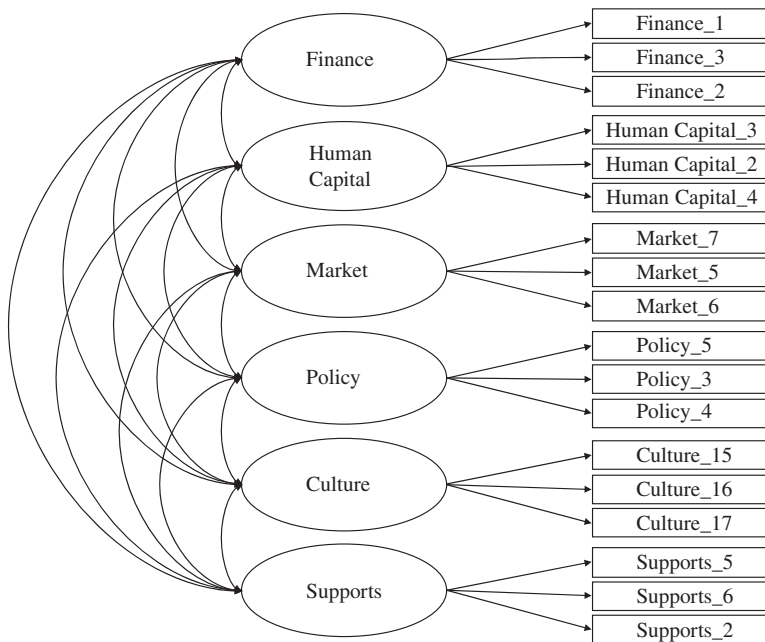
Table 2. Rotated component matrix^a.

	Component					
	1	2	3	4	5	6
Finance_2	0.79					
Finance_4	0.73					
Finance_5	0.71					
Finance_1	0.71					
Finance_3	0.65					
Supports_3		0.74				
Supports_4		0.71				
Supports_2		0.69				
Supports_5		0.64				
Supports_6		0.59				
Culture_17			0.79			
Culture_16			0.79			
Culture_15			0.68			
Human Capital_3				0.8		
Human Capital_2				0.76		
Human Capital_4				0.71		
Markets_6					0.8	
Markets_5					0.77	
Markets_7					0.69	
Policy_3						0.79
Policy_4						0.77
Policy_5						0.49

Extraction Method: Common Factor

Rotation Method: Orthogonal

a. Rotation converged in seven iterations; Loadings below .4 are not displayed

All factor loadings are significant at $P < .05$; $N = 198$ Figure 1.** The final CFA structure of the MEES.

Although, our CFA model did not pass the traditional chi-square goodness-of-fit test, the more commonly applied test of fit is the normed-chi-square goodness-of-fit measure. The critical cut off value for the normed-chi-square goodness-of-fit test is three and lower (Kline 2005). As shown in

Table 3. Summary of fit statistics.

	Sample 2 (Entrepreneur Sample)
Sample Size	267
Number of Parameters	69
DF	120
Chi-Sq Goodness-of-Fit	292.99
Chi-Sq P_Value	0.00
Normed Chi-Sq	2.44
CFI	0.90
RMSEA	0.07

Table 3 our data sample and model passes the normed chi-square goodness-of-fit and model parsimony test. Furthermore, the accepted threshold for the CFI is .9 (Bentler 1990). Hence, our CFA model passed the CFI test of incremental fit, as we achieved a CFI value of .9. Finally, the RMSEA value for the CFA model fell below the traditional .08 cut-off value (MacCallum, Browne, Sugawara, 1996). Thus, the CFA model was able to pass all of the model fit tests except for the chi-square goodness-of-fit test, which is most likely a function of our large sample size. In sum, the data fit the model based on the results of the commonly accepted fit statistics.

In addition to displaying acceptable fit statistics, the model also displayed a desirable set of convergent validity statistics. That is, all of the loadings were greater than the traditional minimum cut-off of .5 and most of the loadings were greater than .7, also all of the loadings were statistically significant (Hair et al. 2006). Moreover, each of the factors within the overall MEES displays excellent internal convergent validity properties in terms of composite reliability. Each of the factors achieves composite reliability coefficients of greater than the standard .7 cut-off value (Bagozzi and Yi 1988; Nunnally 1978), and range from .79 to .73 with an average reliability of .77. Furthermore, the robustness of this models convergent validity is demonstrated by the average variance extracted (AVE) coefficients that fall largely above the cut-off value of .5 (Bagozzi and Yi 1988), as the AVEs for each factor ranged between .49 and .57 with an average value of .53. Accordingly, the items of the MEES on average explained the majority of the variance within each factor. See **Table 4** for a summary of the loadings and convergent validity statistics.

Table 4. Summary of loadings and convergent validity statistics.

	Loadings	P	Reliability	AVE
Finance Items			0.79	0.49
Finance_1	0.64	***		
Finance_3	0.74	***		
Finance_2	0.86	***		
Human Capital Items			0.75	0.56
Human Capital_4	0.62	***		
Human Capital_2	0.81	***		
Human Capital_3	0.68	***		
Market Items			0.77	0.55
Markets_7	0.63	***		
Markets_5	0.78	***		
Markets_6	0.76	***		
Policy Items			0.78	0.53
Policy_5	0.59	***		
Policy_3	0.75	***		
Policy_4	0.86	***		
Culture Items			0.79	0.50
Culture_15	0.61	***		
Culture_16	0.84	***		
Culture_17	0.78	***		
Support Items			0.73	0.57
Supports_5	0.65	***		
Supports_6	0.78	***		
Supports_2	0.64	***		

Table 5. Summary of correlations and squared correlations between factors.

	Culture	Policy	Finance	Supports	Human Capital	Markets
Culture	-	0.27	0.11	0.24	0.08	0.13
Policy	0.52	-	0.09	0.27	0.13	0.13
Finance	0.33	0.30	-	0.37	0.26	0.21
Supports	0.49	0.52	0.61	-	0.40	0.38
Human Capital	0.27	0.36	0.51	0.63	-	0.38
Markets	0.36	0.36	0.46	0.62	0.62	-

*Standard bivariate correlations are listed below the diagonal, squared correlations are listed above the diagonal

**All correlations between factors are significant $P < .05$

To provide further support for the construct validity of the MEES, we also tested the discriminate validity properties of the MEES. To provide a test of the discriminant validity of the measure, we applied the conservative test of comparing each factor's AVE statistic to the squared correlations between each of the factors (see Table 5 for a summary of correlations and squared correlations between factors) (Farrell 2010; Fornell and Larcker 1981). In all cases, the AVEs for each factor were greater than the squared correlations between the factors, as the squared correlations ranged between .09 and .40. Our results suggest that although the factors within the MEES are related to one another, the items that comprise each factor tap a distinct component of the entrepreneurial ecosystem.

In sum, the CFA yielded a set of results that were supportive of the MEES across a number of dimensions. First, the MEES displayed a set of adequate fit statistics; the CFA model represented the data very well. Second, the model demonstrated convergent validity, as each of the items representing the factors significantly loaded on to their corresponding factor. Moreover, the standardized loadings were also large enough to represent the factors on a meaningful level (i.e. loadings ranged from .59–.86) (Hair et al. 2006). The MEES was able to also achieve acceptable convergent validity statistics in terms of both reliabilities and AVEs (Bagozzi and Yi 1988). Third, each of the factors demonstrated a high degree of discriminant validity, as the AVEs for each factor were all greater than the squared correlations between factors (Fornell and Larcker 1981). Therefore, the triangulation of the evidence of model fit, convergent validity, and discriminant validity serve to establish the overall construct validity of the MEES.

Discussion

We began by inquiring about drivers of entrepreneurship and what can make a region more or less suitable for sustaining an entrepreneurial cluster. While current studies and literature have added value to this topic, the applicability is disconnected such that it is difficult to compare varying entrepreneurial ecosystems through the use of the current measures. We set out to create a measure that can be utilized across levels as a means to help solve this problem given its large-scale importance to overall economic conditions. To do so, we grounded the MEES in Isenberg's (2010a, 2011a, 2011b) conceptual work and push it forward by systematizing the six domains of an entrepreneurship ecosystem by determining individuals' perceptions of their entrepreneurship communities. As such, we confirm Isenberg's conceptual model (2010a, 2011a, 2011b) by illustrating a stable factor structure in the MEES. That is, each factor of the MEES captures a unique and important piece of any given entrepreneurial system.

We further contribute to the entrepreneurship ecosystem literature by demonstrating ecosystems can be measured through perceptual techniques. As a result of our research effort, we built a psychometrically robust measure to capture entrepreneurs' perceptions of the surrounding ecosystem. As stated previously, we designed the MEES to be a measure capable of capturing perceptions of the ecosystem of on many levels (i.e. city, state, region, etc.) provided the researchers are able to draw from a representative sample. Going forward this is a tool that not only has academic implications, but also practical value. More specifically, by knowing which domains

within a given ecosystem are strengths and weaknesses, policy makers can gain richer and deeper insight to the fundamental dynamics driving the success of their community ecosystem. This knowledge can lead to more targeted and informed interventions, thus yielding more promising outcomes. Given the decline in the number of start-ups in the United States over the last few decades coupled with the known impact of entrepreneurship on the U.S. economy (Clifton and Badal 2014), any tool that empowers policy makers to improve their ecosystems is much welcomed.

Limitations and future research directions

Our study is not without limitations. First, our data was collected from various areas of the United States, including the West, South, Midwest, and Southeast, but not the Northeast. While this could minimally limit the generalizability of the MEES, *post hoc* analyses of differences between the regions from which data was collected did not reveal differences that would meaningfully suggest the measurement is regionally biased within the United States based sample. Future research, however, is needed to test the generalizability of the MEES outside of the United States.

A second potential limitation is a lack of depth within each domain. It is possible to continue to break down the six domains even further. One example could be breaking the culture domain into two smaller domains (viz. cultural norms and narratives), both of which are known to be relevant cultural factors (Mack and Mayer 2016; Isenberg 2010a). That said, the MEES is a tool designed to help researchers and policy makers better and more quickly assess the areas of strengths and weaknesses within an ecosystem from the perception of the community. The tool is designed to enable these stakeholders to look at the domains and determine which are strong and which are weak, so tactically a weak domain can then be explored in more depth to determine how it can be strengthened. It is not designed to explore a given domain in great detail, so while it is possible to break it down even further, our work stops short of diving into each domain. This limitation is reasonable given ‘academic research can rarely deliver fully developed solutions to any practical problem’ (Davidsson 2009, 213). Future work should explore measurement of each of these domains in greater detail.

In closing the MEES is a step towards better understanding what makes any geographical area more or less conducive towards entrepreneurship. We set out to develop a measure that could be used to better and more quickly assess the strengths and weaknesses of a given ecosystem using the domains posited by the Babson Entrepreneurial Ecosystem Project (e.g. Isenberg 2010a, 2011b). What resulted is a measure, the MEES, which does just that, and several economic development organizations have begun to look to use the MEES in their regions (Chicago, Philadelphia, Tampa). Despite the acknowledged limitations, our hope is that scholars will continue to explore more and better ways to measure entrepreneurial ecosystem dynamics, and that policy makers begin to make more data-driven decisions as it related to fostering entrepreneurship in their regions.

Notes

1. At least three items were always retained because the ultimate goal was to use the items in a confirmatory factor analysis, and three items are required in order achieve model identification for each factor within a confirmatory factor analysis.
2. The model was specified to settle upon six-factor solution.

Disclosure statement

No potential conflict of interest was reported by the authors.

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